Tutorial 3:
practice & examples

A classification problem using SLP

- Assume you have a perceptron with two inputs denoted by x_1 and x_2, and one threshold corresponding to a fixed input denoted by x_0=-1. The perceptron is based on a Signum activation function of output values +1 (for non-negative input only) and -1.

- Starting with the initial weights w_1=+1.0, w_2=+1.0, and an initial threshold w_0=-1.5, calculate all the steps and weight adjustments involved with its training to learn a two-input OR problem.

- To achieve this, use the Perceptron training algorithm with a learning rate η=0.5.

- During training, in each epoch, present the input patterns in the following order: (0,0), (0,1), (1,0) and (1,1). Also, map the "false" (or 0) output to class label -1 and the "true" to +1.

Solution:

⌘ First write the Perceptron model for its output:

$$y = \phi\left[\mathbf{w}^T\mathbf{x}\right] = \phi\left[\sum_{i=0}^{2} x_i w_i\right], \quad \text{where}$$

$$x_0 = -1 \text{ (fixed)}$$

$$\phi[v] = \begin{cases} +1 & v \geq 0 \\ -1 & v < 0 \end{cases}$$

⌘ Then, we write down the complete patterns for the OR problem, and the desired response d (converted to -1 and +1):

Pattern	x_0	x_1	x_2	OR	d
1	-1	0	0	0	-1
2	-1	0	1	1	+1
3	-1	1	0	1	+1
4	-1	1	1	1	+1

⌘ Then, we apply the training algorithm, pattern-by-pattern until no changes are possible:

$$\mathbf{w}(t+1) = \mathbf{w}(t) + \eta\left[d(t) - y(t)\right]\mathbf{x}(t)$$

$$d(t) = \begin{cases} +1 & \text{if } \mathbf{x}(t) \text{ belongs to } C_1 \\ -1 & \text{if } \mathbf{x}(t) \text{ belongs to } C_2 \end{cases}$$

Epoch 1

Currently weight vector is: **w**=[-1.5, 1, 1] (from initial values)

- **Process pattern 1, x=[-1,0,0], d=-1.**
- y=ϕ(+1.5)=+1 ➔ misclassified, so change weights to:
- [-1.5,1,1] + 0.5 × (d-y) × [-1,0,0]= [-1.5,1,1] + -1 × [-1,0,0]= [-0.5,1,1]

Currently weight vector is: **w**=[-0.5, 1, 1] (1st weight update)

- **Process pattern 2, x=[-1,0,1], d=+1.**
- y=ϕ(+1.5)=+1 ➔ correctly classified, so nothing to change!
- [-0.5,1,1] + 0.5 × 0 × [-1,0,1]= same as before.

- **Process pattern 3, x=[-1,1,0], d=+1.**
- y=ϕ(+1.5)=+1 ➔ correctly classified, so nothing to change!
- [-0.5,1,1] + 0.5 × 0 × [-1,1,0]= same as before.

- **Process pattern 4, x=[-1,1,1], d=+1.**
- y=ϕ(+2.5)=+1 ➔ correctly classified, so nothing to change!
- [-0.5,1,1] + 0.5 × 0 × [-1,1,1]= same as before.

Epoch 2

Currently weight vector is: **w**=[-0.5, 1, 1] (from previous 1st update)

- **Process pattern 1, x=[-1,0,0], d=-1.**
- y=ϕ(+0.5)=+1 ➔ misclassified, so change weights to:
- [-0.5,1,1] + 0.5 × (d-y) × [-1,0,0]= [-0.5,1,1] + -1 × [-1,0,0]= [0.5,1,1]

Currently weight vector is: **w**=[+0.5, 1, 1] (2nd weight update)

- **Process pattern 2, x=[-1,0,1], d=+1.**
- y=ϕ(+0.5)=+1 ➔ correctly classified, so nothing to change!
- [0.5,1,1] + 0.5 × 0 × [-1,0,1]= same as before.

- **Process pattern 3, x=[-1,1,0], d=+1.**
- y=ϕ(+0.5)=+1 ➔ correctly classified, so nothing to change!
- [0.5,1,1] + 0.5 × 0 × [-1,1,0]= same as before.

- **Process pattern 4, x=[-1,1,1], d=+1.**
- y=ϕ(+1.5)=+1 ➔ correctly classified, so nothing to change!
- [0.5,1,1] + 0.5 × 0 × [-1,1,1]= same as before.

Epoch 3

Currently weight vector is: **w**=[0.5, 1, 1] (from previous 2nd update)

- Here we can try all patterns once more, in order to verify that all patters are correctly classified.
- We can do that, by trying one-by-one patterns as before, or using matrix calculations as:

$$X = \begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$w = \begin{bmatrix} 0.5 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Perceptron output} = \phi(\mathbf{Xw}) = \phi\left(\begin{bmatrix} -0.5 \\ 0.5 \\ 0.5 \\ 1.5 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ +1 \\ +1 \\ +1 \end{bmatrix} = \text{Desired output}$$